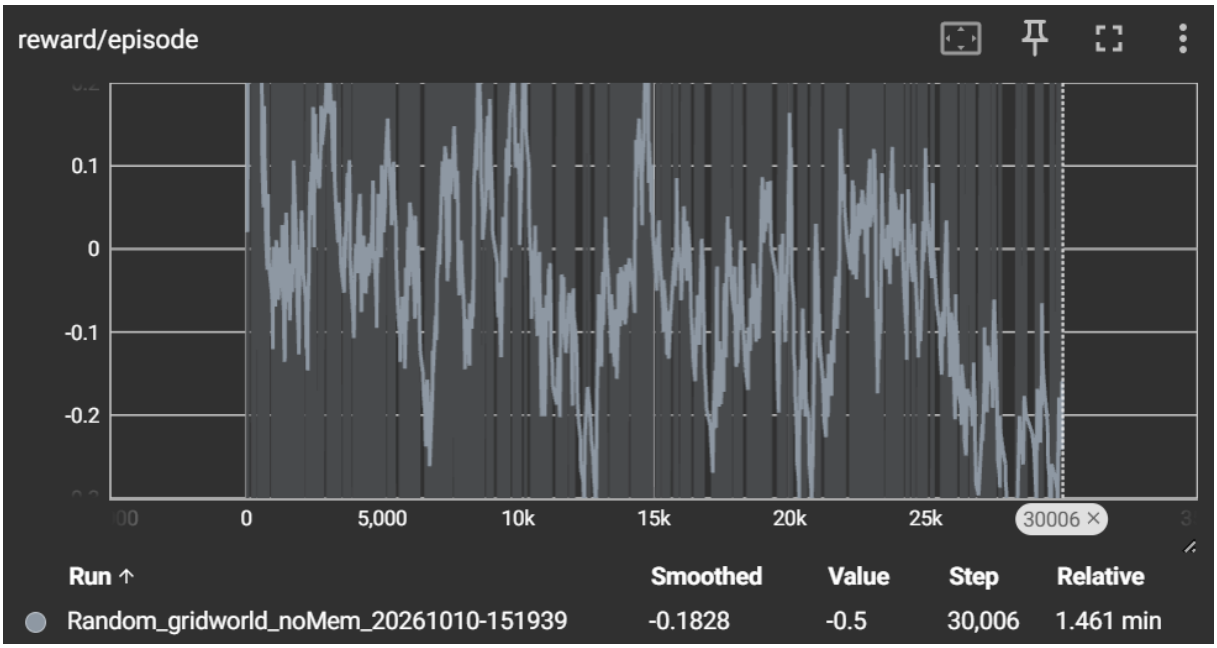
1. Test description

In the following 3 models are tested against each other. All of them are deep reinforcement learning models trying to solve randomly created 8x8 Gridworld problems with random start and goal positions. All models see the agent's current position and the 4 tiles surrounding them. The first model has no memory, therefore stumbling in the dark about where it already has been. The second model knows for the 4 surrounding tiles how often they have already been visited. The third model has an LSTM part integrated into its agent, allowing it to learn useful knowledge on its own.

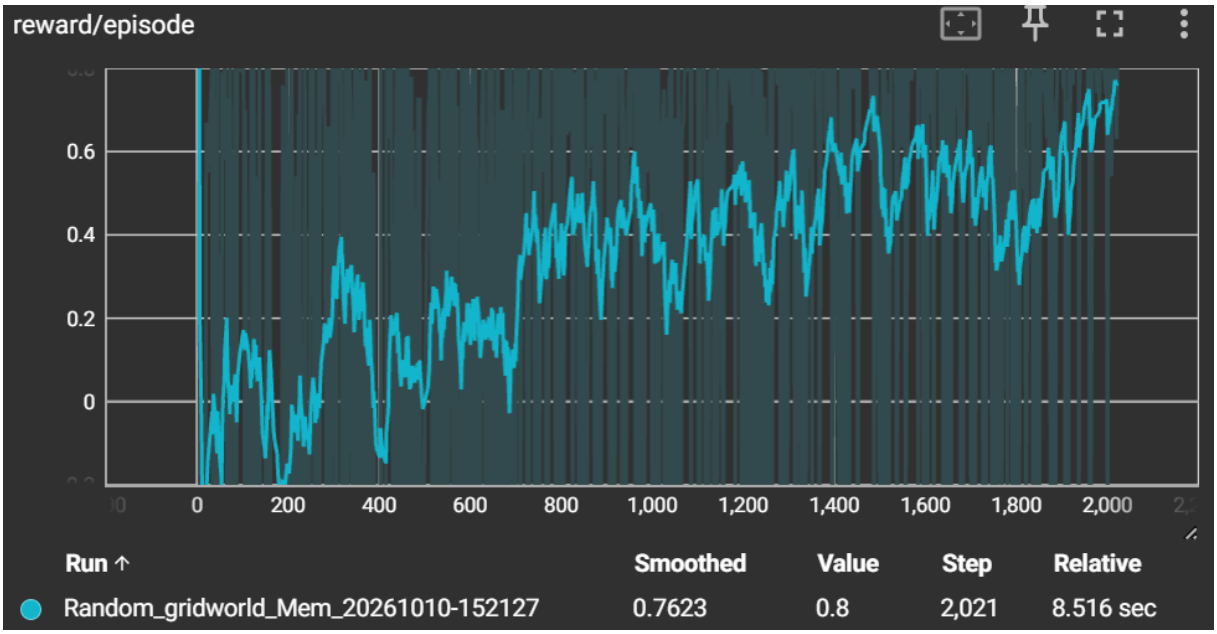
2. Results

Below are the smoothed reward values per episode depicted for each model. The agent gets a reward of 1 for finding the goal and a reward of -0.01 for each step taken, forcing it to find the goal as fast as possible.

No memory:



Handmade memory:



LSTM:



3. Analysis

We see that the model without memory has no or even a negative learning curve, scoring around 0 reward each episode. The handcrafted memory performs the best, solving the environment in around 2000 episodes (a reward of 0.7 is needed to solve the episode, the theoretical maximum reward is 1). The LSTM also solves the environment but only after around 30000 episodes, therefore performing far worse than the former mentioned model.